

Chi Square Test

Question

- Suppose we have seq of 4000 3 digit number (000 to 999)
- and if can expect about 400 members in each range the observed value is given as in table and significance level is 1 %:

00-99	425
100-199	378
200-299	395
300-399	415
400-499	420
500-599	395
600-699	430
700-799	383
800-899	365
900-999	394

Step 1: Identify Variables

Total Observations $N = 4000$ (Given in Question)

No of Classes $n = 10$ (Identify from Question)

Expected Observation $E_i = N/n = 4000/10 = 400$

$u = n - 1 = 10 - 1 = 9$

Step 2 : To calculate $\{(O_i - E_i)^2\}/E_i$

Intervals	E_i	O_i	$O_i - E_i$	$(O_i - E_i)^2$	$\{(O_i - E_i)^2\}/E_i$
00-99	400	425	25	625	1.5625
100-199	400	378	-22	484	1.21
200-299	400	395	-5	25	0.0625
300-399	400	415	15	225	0.5625
400-499	400	420	20	400	1
500-599	400	395	-5	25	0.0625
600-699	400	430	30	900	2.25
700-799	400	383	-17	289	0.7225
800-899	400	365	-35	1225	3.0625
900-999	400	394	-6	36	0.09

Step 3: Calculate χ^2

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

$\chi^2 = 10.585$ (Calculated from the above table)

Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67
10	2.558	3.940	4.865	6.737	9.342	12.55	15.99	18.31	23.21
11	3.053	4.575	5.578	7.584	10.341	13.70	17.28	19.68	24.72
12	3.571	5.226	6.304	8.438	11.340	14.85	18.55	21.03	26.22
13	4.107	5.892	7.042	9.299	12.340	15.98	19.81	22.36	27.69
14	4.660	6.571	7.790	10.165	13.339	17.12	21.06	23.68	29.14
15	5.229	7.261	8.547	11.037	14.339	18.25	22.31	25.00	30.58
16	5.812	7.962	9.312	11.912	15.338	19.37	23.54	26.30	32.00
17	6.408	8.672	10.085	12.792	16.338	20.49	24.77	27.59	33.41
18	7.015	9.390	10.865	13.675	17.338	21.60	25.99	28.87	34.80
19	7.633	10.117	11.651	14.562	18.338	22.72	27.20	30.14	36.19
20	8.260	10.851	12.443	15.452	19.337	23.83	28.41	31.41	37.57
22	9.542	12.338	14.041	17.240	21.337	26.04	30.81	33.92	40.29
24	10.856	13.848	15.659	19.037	23.337	28.24	33.20	36.42	42.98
26	12.198	15.379	17.292	20.843	25.336	30.43	35.56	38.89	45.64
28	13.565	16.928	18.939	22.657	27.336	32.62	37.92	41.34	48.28
30	14.953	18.493	20.599	24.478	29.336	34.80	40.26	43.77	50.89
40	22.164	26.509	29.051	33.660	39.335	45.62	51.80	55.76	63.69
50	27.707	34.764	37.689	42.942	49.335	56.33	63.17	67.50	76.15
60	37.485	43.188	46.459	52.294	59.335	66.98	74.40	79.08	88.38

Step 4: Calculate $X_{u,\alpha}^2$ from Distribution table

$$X_{u,\alpha}^2 = 21.67$$

Step 5 : Condition

If $X^2 \geq X_{u,\alpha}^2$ Then Null Hypothesis is rejected and Random numbers is not in uniform distribution

$$X^2 = 10.585$$
$$X_{u,\alpha}^2 = 21.67$$

For the above values $X^2 < X_{u,\alpha}^2$; So, Null Hypothesis is accepted and Random numbers is in uniform distribution